

source home automation platform that emphasises flexibility and customisation. It provides a powerful environment for integrating and automating smart devices.

Code example - OpenHAB rules (DSL-based)

```
// Example: Turn on the heating when
the temperature drops below 20°C
rule "Temperature Control"
when
    Item temperature_sensor changed
then
    if (temperature_sensor.state < 20)
    {
        heating_switch.sendCommand(ON)
    } else {
        heating_switch.sendCommand(OFF)
    }
end
```

OpenHAB uses a domain-specific language (DSL) for creating automation rules. In this example, the rule monitors the temperature sensor and controls the heating system accordingly.

Node-RED

Node-RED is a flow-based development tool for visual programming, ideal for creating IoT applications with a user-friendly approach. It simplifies the integration of devices, APIs, and online services. Here's what makes Node-RED stand out.

Code example - Node-RED flow for visual programming

```
[
  {
    "id": "1",
    "type": "inject",
    "payload": "{ \"temperature\": 22.5 }",
    "repeat": "10",
    "crontab": "",
    "once": false,
    "x": 150,
    "y": 220,
    "wires": [

```

```
      "2"
    ]
  },
  {
    "id": "2",
    "type": "ui_gauge",
    "name": "Temperature",
    "group": "1",
    "order": 0,
    "width": 0,
    "height": 0,
    "gtype": "gauge",
    "title": "Temperature",
    "label": "°C",
    "format": "{{value}}",
    "min": 0,
    "max": "40",
    "colors": [
      "#00b500",
      "#e6e600",
      "#ca3838"
    ],
    "x": 350,
    "y": 220,
    "wires": [ ]
  },
  {
    "id": "3",
    "type": "ui_group",
    "name": "Sensor Data",
    "order": 1,
    "disp": true,
    "width": "6",
    "collapse": false
  },
  {
    "id": "4",
    "type": "ui_tab",
    "name": "Dashboard",
    "icon": "dashboard",
    "order": 1,
    "disabled": false,
    "hidden": false
  }
]
```

This shortened Node-RED flow simulates temperature data and displays it on a gauge in the Node-RED dashboard. It maintains the core functionality while reducing the code for brevity.

Benefits of open source in IoT

Open source is not just a development model; it's a philosophy that has significant advantages for IoT projects. In this section, we'll explore the key benefits of incorporating open source in the realm of IoT development.

Cost-effectiveness

■ Reduced development costs

Building IoT solutions from the ground up can be a resource-intensive endeavour. Open source software eliminates the need to start from scratch, saving both time and money. Developers can leverage existing open source components and frameworks to accelerate development.

■ No licensing fees

One of the most appealing aspects of open source is the absence of licensing fees. Traditional proprietary software often comes with substantial licensing costs. In contrast, open source allows organisations to use, modify, and distribute software freely, significantly reducing the financial burden on IoT projects.

Flexibility and customisation

■ Tailoring solutions to specific needs

Open source empowers developers to tailor IoT solutions to meet specific requirements. Whether it's customising features, interfaces, or integrations, open source code can be modified to fit unique use cases, ensuring a precise fit for your project.

■ Avoiding vendor lock-in

Proprietary solutions often result in vendor lock-in, making it challenging to migrate or adapt when needs change. Open source liberates organisations from such constraints, allowing for long-term flexibility and control over technology choices.

Community collaboration

■ Global developer community

Open source thrives on community collaboration. A diverse and global community of developers contributes